Practice 2 ALL — Unified Enterprise Operations, SDP, Delta & Lakeview Suite

Taskwise screenshot submission template

- **Trainee name**: _____
- **Candidate ID**: _____
- **Existing catalog**: _____
- **Date**: _____
- **Schema**: `practice_2` | **Volumes**: `p2_all_ext_files` (External), `taskv_yourname` (Managed)

Submit only the named taskwise screenshots in the matching template slots below. Code and output have separate slots whenever both are required. Capture existing results; do not rerun mutations (`MERGE`, `UPDATE`, `INSERT`, `DELETE`, `OPTIMIZE`) for screenshots.

Task and screenshot allocation

Task	Title	Marks	Images
1	Prepare External and Managed Unity Catalog Volumes	6	1
2	Build Batch SDP Materialized Views with Quality Expectations	12	2
3	Build Streaming Ingestion & SCD Type 1 CDC in SDP, Execute & Export	14	2
4	Create Managed Delta Ledger, Enforce Schema & Evolve Schema	10	2
5	Synchronize Records via MERGE and Apply Targeted DML	12	2

Task	Title	Marks	Images
6	Inspect Delta Change Data Feed (CDF), Time Travel & Operation History	12	2
7	Run Delta Layout Optimization (ZORDER) & Prove Equivalence	10	2
8	Apply Unity Catalog Column Masking on Contact Phone Numbers	8	2
9	Execute Comprehensive SQL Validation & Rollup Summaries	6	1
10	Build Interactive Lakeview Dashboard & AI/BI Genie Space	10	2

Total required screenshots: 18. Each following section identifies the exact task, filename, and visible evidence for its image.

Task 1 image A — Setup UI

Prepare External and Managed Unity Catalog Volumes
`P2_ALL_T01_A.png`
Volume view showing catalog, `practice_2`, `p2_all_ext_files`, `EXTERNAL` type, assigned ADLS child URI, uploaded `P2_ALL_Master_Enterprise_Dataset.csv` inside `input/`, and `dbutils.fs.ls()` confirmation.
This is image 1 of 1 required for Task 1.
Expected result: Both volumes exist and `P2_ALL_Master_Enterprise_Dataset.csv` is inside `p2_all_ext_files/input/`.

INSERT THE SCREENSHOT FOR `P2_ALL_T01_A.png` HERE
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Task 2 image A — Bronze code

Build Batch SDP Materialized Views with Quality Expectations
`P2_ALL_T02_A.png`
Complete `@dp.materialized_view` definition for `sdp1_bronze_batch`, including imports

(`from pyspark import pipelines as dp`), volume path, and explicit 14-column schema.
This is image 1 of 2 required for Task 2.
Expected result: Bronze keeps all 10 raw rows and 14 columns with `event_date` as `STRING`.

INSERT THE SCREENSHOT FOR `P2_ALL_T02_A.png` HERE
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Task 2 image B — Silver code

Build Batch SDP Materialized Views with Quality Expectations
`P2_ALL_T02_B.png`
Complete `@dp.materialized_view` definition for `sdp2_silver_clean` showing `spark.read.table("sdp1_bronze_batch")`, `@dp.expect_or_drop("valid_unit_price", ...)`, `F.trim`/`F.upper` on `city_gallery`, `F.to_date` on `event_date`, `F.coalesce` on `concession_surcharge`, and `net_amount` calculation.
This is image 2 of 2 required for Task 2.
Expected result: Silver drops `REC104` and `REC106` and transforms the remaining 8 valid records.

INSERT THE SCREENSHOT FOR `P2_ALL_T02_B.png` HERE
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Task 3 image A — CDC & Export code

Build Streaming Ingestion & SCD Type 1 CDC in SDP, Execute & Export
`P2_ALL_T03_A.png`
Complete `@dp.streaming_table` (`p2_all_claim_events`), `dp.create_streaming_table("p2_all_current_entities")`, `dp.apply_changes` (key `entity_id`, sequence `change_sequence`, SCD Type 1), and separate notebook `.write.mode("overwrite").option("header", "true").csv(...)` export code.
This is image 1 of 2 required for Task 3.
Expected result: Streaming ingestion, keyed Type 1 CDC flow, and CSV export to managed volume are properly defined.

INSERT THE SCREENSHOT FOR `P2_ALL_T03_A.png` HERE
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Task 3 image B — Pipeline result

Build Streaming Ingestion & SCD Type 1 CDC in SDP, Execute & Export
`P2_ALL_T03_B.png`
Completed `p2_all_enterprise_pipeline` graph showing destination `<your_catalog>.practice_2`, `sdp1_bronze_batch` (10), `sdp2_silver_clean` (8 saved, 2 dropped in quality panel), `p2_all_claim_events` (10), and `p2_all_current_entities` (7).
This is image 2 of 2 required for Task 3.
Expected result: All 4 pipeline tables complete successfully with accurate row and dropped counts.

INSERT THE SCREENSHOT FOR `P2_ALL_T03_B.png` HERE
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Task 4 image A — Error output

Create Managed Delta Ledger, Enforce Schema & Evolve Schema
`P2_ALL_T04_A.png`
Readable schema-mismatch error when attempting to append `REC109` and `REC110` (with `extra_grade`) to `p2_all_operations_ledger` without schema evolution enabled.
This is image 1 of 2 required for Task 4.
Expected result: Delta schema enforcement rejects the unexpected `extra_grade` column.

INSERT THE SCREENSHOT FOR `P2_ALL_T04_A.png` HERE
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Task 4 image B — Evolved output

Create Managed Delta Ledger, Enforce Schema & Evolve Schema
`P2_ALL_T04_B.png`
Code showing `delta.enableChangeDataFeed = true` and `.option("mergeSchema", "true")`, followed by all 8 saved rows in `p2_all_operations_ledger` (`NULL` `extra_grade` for `REC101`–`REC108`, `GRADE_A`/`GRADE_B` for `REC109`–`REC110`) and recorded `baseline_version`.
This is image 2 of 2 required for Task 4.

Expected result: 8 rows saved with evolved `extra_grade` column and captured `baseline_version`.

INSERT THE SCREENSHOT FOR `P2_ALL_T04_B.png` HERE
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Task 5 image A — MERGE & DML code

Synchronize Records via MERGE and Apply Targeted DML
`P2_ALL_T05_A.png`
Complete `MERGE INTO p2_all_operations_ledger` (`REC103` update + `REC111` insert), targeted `UPDATE` on `REC102` (recording `update_version`), `INSERT` of `REC112`, and `DELETE` of `REC105` (recording `delete_version`).
This is image 1 of 2 required for Task 5.
Expected result: All four Delta mutations and version captures are clearly visible.

INSERT THE SCREENSHOT FOR `P2_ALL_T05_A.png` HERE
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Task 5 image B — Current ledger output

Synchronize Records via MERGE and Apply Targeted DML
`P2_ALL_T05_B.png`
Printed `update_version` and `delete_version`, plus all 9 current rows in `p2_all_operations_ledger` ordered by `record_id` (`REC105` absent).
This is image 2 of 2 required for Task 5.
Expected result: 9 active rows with updated `REC102` (`1250.00`), updated `REC103` (`1230.00`), and inserted `REC111`/`REC112`.

INSERT THE SCREENSHOT FOR `P2_ALL_T05_B.png` HERE
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Task 6 image A — CDF output

Inspect Delta Change Data Feed (CDF), Time Travel & Operation History
`P2_ALL_T06_A.png`
CDF output bounded from `update_version` through `delete_version` showing

`record_id`, `unit_price_fee`, `net_amount`, `event_status`, `_change_type`, `_commit_version`, and `_commit_timestamp` for the 4 change rows (`REC102` pre/post, `REC112` insert, `REC105` delete).
This is image 1 of 2 required for Task 6.
Expected result: Exactly 4 CDF events displayed with real commit metadata; baseline inserts excluded.

INSERT THE SCREENSHOT FOR `P2_ALL_T06_A.png` HERE
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Task 6 image B — Time Travel & History output

Inspect Delta Change Data Feed (CDF), Time Travel & Operation History
`P2_ALL_T06_B.png`
`VERSION AS OF <baseline_version>` vs. `Current` summary counts/totals (`8 / 7355.00` vs `9 / 8245.00`), side-by-side `REC102` and `REC105` comparison, and `DESCRIBE HISTORY` showing `MERGE`, `UPDATE`, `WRITE`, and `DELETE`.
This is image 2 of 2 required for Task 6.
Expected result: Historical state and operation audit history verified.

INSERT THE SCREENSHOT FOR `P2_ALL_T06_B.png` HERE
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Task 7 image A — Optimization & Comparison code

Run Delta Layout Optimization (ZORDER) & Prove Equivalence
`P2_ALL_T07_A.png`
Single `OPTIMIZE p2_all_operations_ledger ZORDER BY (centre_zone)` command, returned metrics, and bidirectional duplicate-preserving comparison code (`EXCEPT ALL` or `.exceptAll()`).
This is image 1 of 2 required for Task 7.
Expected result: `OPTIMIZE ZORDER BY` executes and duplicate-preserving comparison is coded in both directions.

INSERT THE SCREENSHOT FOR `P2_ALL_T07_A.png` HERE
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Task 7 image B — Equivalence output

Run Delta Layout Optimization (ZORDER) & Prove Equivalence
`P2_ALL_T07_B.png`
Displayed difference counts (`Current minus Pre-Opt: 0` and `Pre-Opt minus Current: 0`), current row count (`9`), and the single sentence explaining that physical layout optimization does not alter logical rows.
This is image 2 of 2 required for Task 7.
Expected result: Both difference counts equal `0` with explanation sentence visible.

INSERT THE SCREENSHOT FOR `P2_ALL_T07_B.png` HERE
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Task 8 image A — Mask code & Attachment metadata

Apply Unity Catalog Column Masking on Contact Phone Numbers
`P2_ALL_T08_A.png`
`CREATE FUNCTION p2_all_phone_mask`, `ALTER TABLE p2_all_operations_ledger ALTER COLUMN contact_phone SET MASK p2_all_phone_mask`, and `DESCRIBE TABLE EXTENDED` metadata confirming attachment.
This is image 1 of 2 required for Task 8.
Expected result: Metadata proves `p2_all_phone_mask` is attached to `contact_phone`.

INSERT THE SCREENSHOT FOR `P2_ALL_T08_A.png` HERE
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Task 8 image B — Protected output

Apply Unity Catalog Column Masking on Contact Phone Numbers
`P2_ALL_T08_B.png`
Direct `SELECT record_id, entity_id, contact_phone FROM p2_all_operations_ledger ORDER BY record_id` output showing `******1001`, `******1002`, `NULL` (`REC107`), `******1007`, `******1008`, `******1009`.
This is image 2 of 2 required for Task 8.
Expected result: 8 masked phone numbers and 1 `NULL` value displayed directly from the table.

INSERT THE SCREENSHOT FOR `P2_ALL_T08_B.png` HERE
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Task 9 image A — SQL Summary output

Execute Comprehensive SQL Validation & Rollup Summaries
`P2_ALL_T09_A.png`
SQL output showing `centre_zone` counts and `SUM(net_amount)` (`NORTH_A`: `4 / 3620.00`, `SOUTH_B`: `3 / 3750.00`, `EAST_C`: `2 / 875.00`, `Overall`: `9 / 8245.00`) and daily `event_date` totals (`2026-10-18`: `3470.00`, `2026-10-19`: `2115.00`, `2026-10-20`: `2660.00`).
This is image 1 of 1 required for Task 9.
Expected result: Zone, daily, and overall totals match reference values.

INSERT THE SCREENSHOT FOR `P2_ALL_T09_A.png` HERE
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Task 10 image A — Unfiltered Dashboard

Build Interactive Lakeview Dashboard & AI/BI Genie Space
`P2_ALL_T10_A.png`
Saved Lakeview dashboard `p2_all_performance_dashboard` showing unfiltered Bar chart by `centre_zone` (`3620.00`, `3750.00`, `875.00`), Line chart by `event_date` (`3470.00`, `2115.00`, `2660.00`), and shared `centre_zone` filter cleared to `All`.
This is image 1 of 2 required for Task 10.
Expected result: Unfiltered Lakeview charts and shared filter control are clearly visible.

INSERT THE SCREENSHOT FOR `P2_ALL_T10_A.png` HERE
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Task 10 image B — Filtered Dashboard & Genie

Build Interactive Lakeview Dashboard & AI/BI Genie Space
`P2_ALL_T10_B.png`
Lakeview dashboard filtered to `SOUTH_B` (Bar: `3750.00`; Line: `2480.00` on `2026-10-18` and `1270.00` on `2026-10-20`) alongside the AI/BI Genie Playground natural-language verification of zone totals.

This is image 2 of 2 required for Task 10.
Expected result: Shared filter updates both charts simultaneously and Genie confirms the exact metrics.

INSERT THE SCREENSHOT FOR `P2_ALL_T10_B.png` HERE
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